Course Project: Final Milestone

Introduction of Topic/Problem

This study investigates the relationship between a movie's production budget, and its reception by audiences and critics, to determine if a large production budget is enough to predict success of a movie, or if there are more variables that dictate if a movie will be a hit or a miss. I will utilize various data sources to provide the most detailed information possible. By joining the various data sources, I will be able to determine if a higher budget translates to higher scores from critics, and a larger profit, or if independent films achieve a better return on investment. Project-wise, *I hope to learn* if a movie's budget can actually predict the success of a movie, or if there are multiple variables that need to be taken into account to predict the success of a film. Predictive Analytics-wise, *I hope to learn* better techniques and hone my skills for modeling and evaluation. I think this problem is interesting to see if a high budget truly means the success of a movie. Based on my professional judgment, this isn't true, or production companies would have massive budgets for every movie they make. I think there are numerous inputs to a film, including casting and writing, that have a big impact on the success of a movie. Film-making companies, production companies, and other groups, including independent filmmakers, would be interested in knowing if there was a way to predict the success of the films they are making.

Overview of Data Used

For data sources, there are two that I utilized for this study. The first is The Movie Database (TMDB) 5000 Movie Dataset from Kaggle, where they were positing if companies (like Pixar) had found a formula for movie success. They were looking to see if they could predict which films would be highly rated or a commercial success. This dataset includes the title, budget, production company, revenue, and some critic metrics for each title included. I also utilized the Open Movie Database (OMDb) API RESTful web service, which provides movie information including content and posters for movies. It includes IMDb ID, movie title, and the plot, as well as the type if it's a movie, series, or episode. This also provides Rotten Tomato scores, ratings, votes, and award information. Since it has additional scores, ratings, votes, and awards information, this data can be critical in determining the success of a film.

Risks and Ethical Implications

To ensure data integrity when I cleaned the data, I needed to evaluate records that had '0' budget or revenue or very low budgets or revenue, in case they represented indie films or made-for-TV movies and not just bad data. To ensure total transparency, I documented all assumptions made during my analysis. Since I decided to use the OMDb API, I was careful to strictly adhere to the OMDb API attribution guidelines for usage and utilized my API key. I had applied for, and received, an API key which has a 1,000 daily limit for calls, so it took multiple pulls over multiple days to have a larger dataset and match the titles in the TMDB 5000 Movie Dataset. I decided to then combine the TMDB and OMDb API datasets; since the TMDB data did not include the IMDb ID I need different way to join the title. To combat this issue I decided to use a simple title match across the datasets. For this, I created a *Simplified_Title* column where I

stripped off special characters and extra space and converted the entire string to lowercase to ensure that every title shared the same information. If a title was missing in one of the sets, I dropped that row rather than insert placeholder values. By starting with a 5000 title set, I was able to maintain enough statistical power to drop some of the records and still be able to adequately train and test my chosen models. Since the data is all from public datasets, there are no ethical issues presented since I am not using the data to create my own site and only utilized it for educational purposes. If I was to use the data to generate my own site to mimic the IMDBPro website, that would present an ethical issue. To mitigate any issues when I worked through my analysis, I thoroughly documented all of my choices.

Methods of Analysis

As my focus is on if a film will make more money based on budget and critic scores, I used regression modeling to predict the box office revenue. The goal of this project was to evaluate how critic scores, or how an extra million dollars in the budget increases a film's revenue. Therefore, I will focused on predicting the revenue of a movie based on budget and critic scores.

For regression algorithms, I used Multiple Linear Regression as a baseline model for evaluating linear relationships in the data. This was my first attempt to validate how the budget, votes, and critic metrics help to predict the revenue. I had also planned to try a Decision Tree Regressor in case there of non-linear relationships between variables. This would most likely prove to be a better option if the high budget only yielded a high revenue if the critic rating was above a certain threshold. This would also allow the movie data to be split into branches based

on feature cut-offs. If neither of these provided the results I was looking for, I planned to also try the Random Forest Regressor. Since this builds multiple decision trees and then averages the predictions, it would do a good job handling any missing values or skewed financial data.

Evaluation for Regression models focuses on the error. Therefore, I used the Mean Absolute Error (MAE) to find the average of the absolute difference between the actual and predicted values. This is an easily interpretable metric that makes sense in the real world as it provides how many dollars (or log dollars) the predictions are off by on average. The Root Mean Squared Error (RMSE) is similar to MAE but squares the errors (the Mean Squared Error or MSE) before averaging and then takes the square root. It measures the standard deviation of residuals. This means larger misses are penalized at a higher rate. This should help in ensuring that the model won't make large prediction errors for revenue. The final evaluation criteria is the R-Squared, which helps to determine the proportion of variance in the target variable that is able to be explained by the features chosen. Also called the coefficient of determination, "R-Squared represents the proportion of the variance in the dependent variable which is explained by the linear regression model."(Chugh, 2020). This should help me understand if my features account for, say, 75% of the variance in movie revenue, that I still have 25% that is some unknown random noise. The R-Squared will also provide statistical proof on if the dataset is large enough for modeling. If the R-Squared value is above 0.70+, then it means that the dataset has enough predictive power to explain the variance in movie success.

Exploratory Data Analysis

Data Ingest and Preparation

I utilized the TMDB dataset as my baseline data, so this is where the Exploratory Data Analysis (EDA) phase began. I began by loading the data and getting some initial metrics from the dataset, including the number of rows that included both a $0 budget and a $0 revenue. If both of these conditions are met, I dropped the rows since it is most likely that a movie with a $0 budget *and* $0 revenue is simply missing data and not a true reflection of that movie. This took the initial 4,803-row dataset down to 3,229 rows. I also extracted the primary genre of the movie from a string of genres, using the first listed genre as the primary genre. I then cleaned the release date column to be a true date and also created a new column holding only the release year. This is important if there is a movie that had been remade, such as *A Star is Born,* which had four major releases from 1937, 1954, 1976, and again in 2018.

I then proceeded to drop columns that were not needed as they would not add to the modeling and would simply distract the model, like production companies, spoken language, overview, and others. I kept 9 columns total, including *Title, Genre, Budget, Revenue, Release Year, Popularity, Runtime, Vote Average,* and *Vote Count.* I added a 10th column by creating a *Simplified Title* where I converted the title to lowercase and removed all special characters and extra internal spaces. This would allow for easy matching between the TMDB and OMDb API dataset. The TMDB data does not include the IMDb ID, as the OMDb data does, so the only way I can join the sets is via *Movie Title*. This is the reason I created the *Simplified Title* column, and stripped unnecessary characters, to make joining data cleaner and easier. After all of the updates, the TMDB dataset sat at 3,229 rows.

Using the TMDB dataset as my foundational dataset, I then created a progress file for holding the OMDb data. Since the API key I have only allows 1,000 queries a day, it took a few

days to fetch all the titles. Each day, I re-ran the fetch, adding the new titles to the progress file based on the list of all the titles from my TMDB data. I limited my daily calls to 975 to ensure I was respectful of the 1,000 queries a day for the OMDb API key use agreement. Once I had all of the data, I began with the EDA process.

First, I paired down columns, removing the *Plot, DVD, Actors, Writer, Director,* and other columns that would just add noise to the model. From this dataset, I kept the *Movie Title, Release Year, Released, Genre, Awards, Ratings, Metascore, IMDb Rating, IMDb Votes,* and *Box Office.* I also renamed columns to match the format with my TMDB data. The next focus was on data types and ensuring they were of the correct type for modeling later. This entailed converting the Release Year to a datetime object, converting the *Metacritic Score, IMDb Rating,* and *IMDb Votes* to numbers. I also stripped the dollar sign and commas out of the *Box Office* numbers and converted them to a numeric type.

I then checked for missing values for each column. This showed my *Movie Title* was not missing any data, so I proceeded from here to then extract the *Rotten Tomatoes Score* from the *Ratings*, which was a nested string, and created a new column for the *Rotten Tomatoes Score*. To simplify the *Awards Data*, I created an *Award Status* column and standardized the scores into four categories: *No Awards, Nominated Only, Award Winner,* and *Oscar Winner*. Finally, I created the same *Simplified Title* I had created for my TMDB data to allow for easy matching between the two datasets. With all the data for both data sets, I proceeded to merge the two datasets based on the Simplified Title to create a single dataset to use for modeling. This left me with 2,990 records in my combined data frame that had 0 missing values.

From here, I moved to analysis and visualizations to help me understand the data on a deeper level and to provide insights for later modeling. To ensure data integrity and feasibility checks for the project, I first checked how closely the TMDB and OMDb revenue numbers were since this was my target variable. I found the correlation between them to be 0.9245. TMDB had 0 missing values for revenue, while OMDb showed 178 missing values. Based on how tightly OMDb was to TMDB's revenue, and the fact that TMDB is not missing data, I used TMDB's *Revenue_USD* as my target variable. Next, I turned to the critic scores, of which I had three different critic scores, from the 2 different data sources. A quick check showed that each of the critic scores was missing scores for titles. The *Rotten Tomatoes Score* was missing from 141 rows, *Metacritic Score* was missing for 178 rows, and the *IMDb Rating* from 23 records. I then chose to create a copy of the DataFrame and impute median values for the missing rows. The Multiple Linear Regression will need every value filled out, while the Decision Tree or Random Forest can deal with the missing values later. I wanted to maintain high statistical power and did not want to loose additional data, which is why I chose to impute the median values.

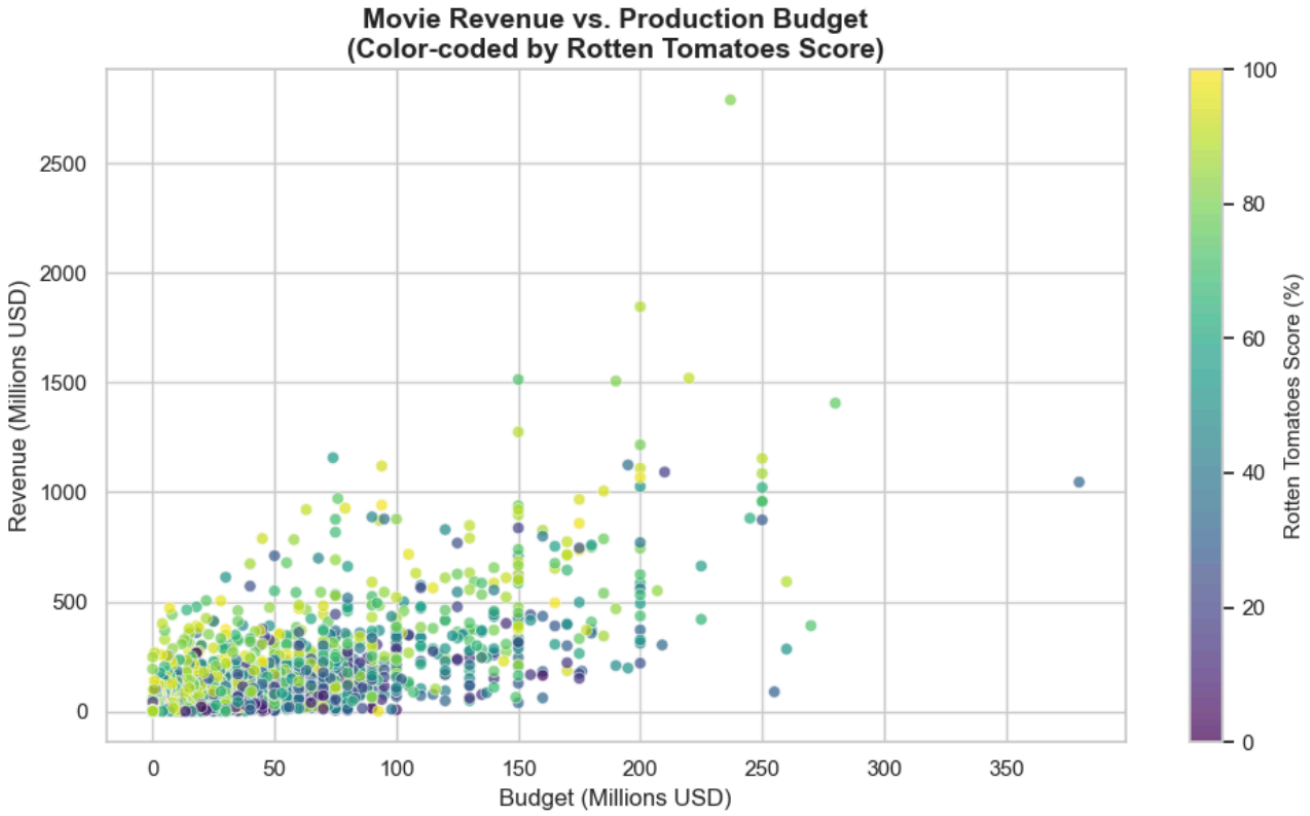
The balance of a dataset is critical to knowing which metrics need to be used to validate models. Class imbalance is more difficult for classification models to handle but can also present a challenge for regression models. The results showed that the data was heavily skewed toward Award Winners at 63.68%. This meant that standard Accuracy was not a good validation metric for a classification model. This is also what led me to focus my efforts on Linear Regression, instead of Classification, since Award Winners and Oscar Winners comprise 73.81% of the data. With the skew of this data, it is likely a classification model would be able to predict a winning

movie but would struggle on the minority and have a difficult time predicting movies that would not win an award.

I then began a series of visualizations to explore the data further. I started with a scatter plot to view my feature versus target variable using the critic score as a gradient. This allowed me to see that there were films with high critic scores and low budgets, and higher budget movies with variation in critic scores. It did appear that as the budget goes up, critic scores are typically higher.

Figure 1

Movie Revenue vs Production Budget

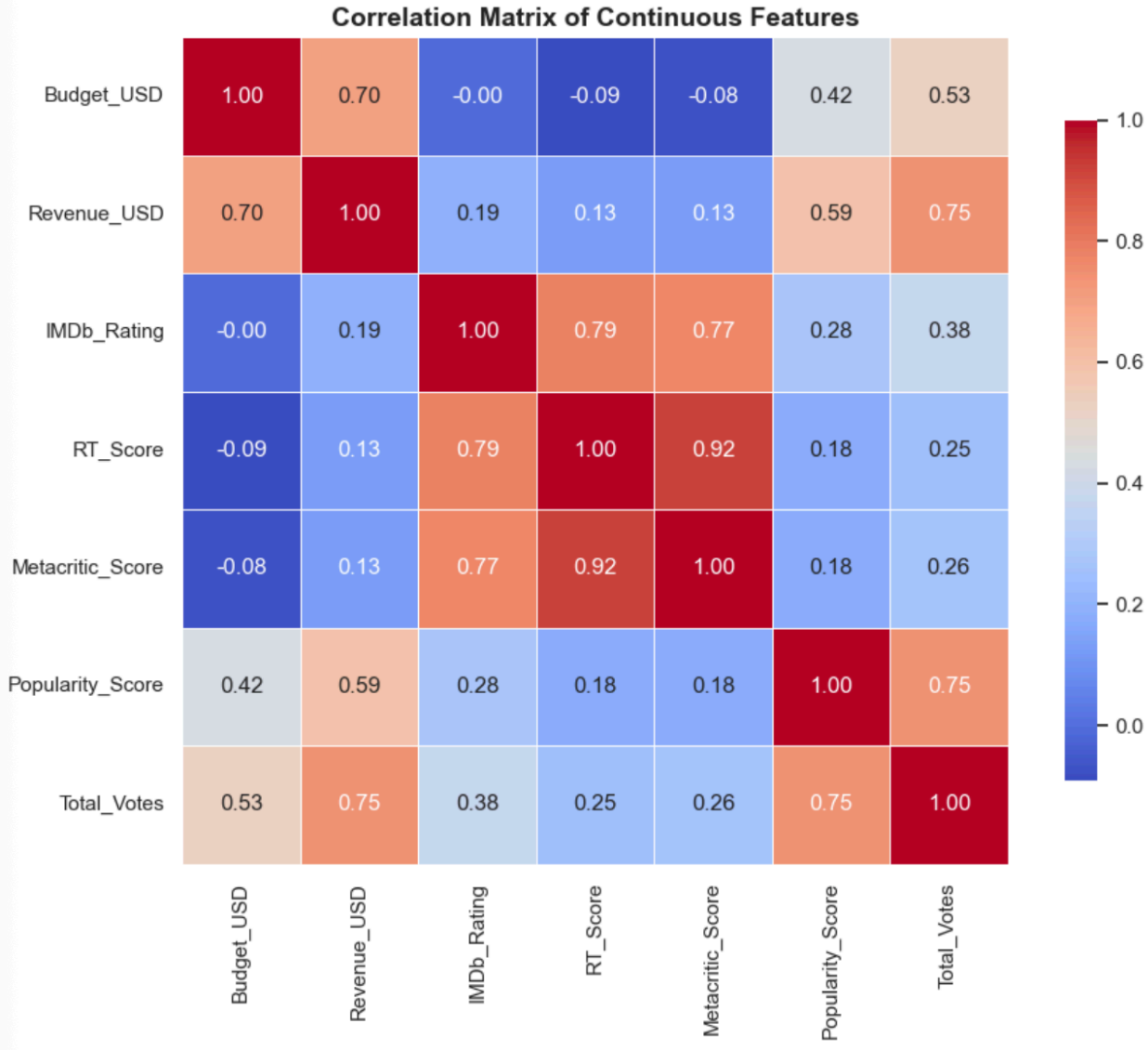


I then assessed the data for multicollinearity by creating a Correlation Heatmap. I specifically focused the matrix to the *Budget, Revenue, IMDb Rating, Rotten Tomatoes Score,*

Metacritic Score, and *Popularity Score*. This showed that the *Metacritic Score* and the *Rotten Tomatoes Score* were tracking incredibly close at 0.92. If both of these were used in a Multiple Linear Regression model, there would be issues due to the high coefficient variance. The choice here would be to either drop one of these metrics or engineer a new metric by averaging these scores together into a single composite score.

Figure 2

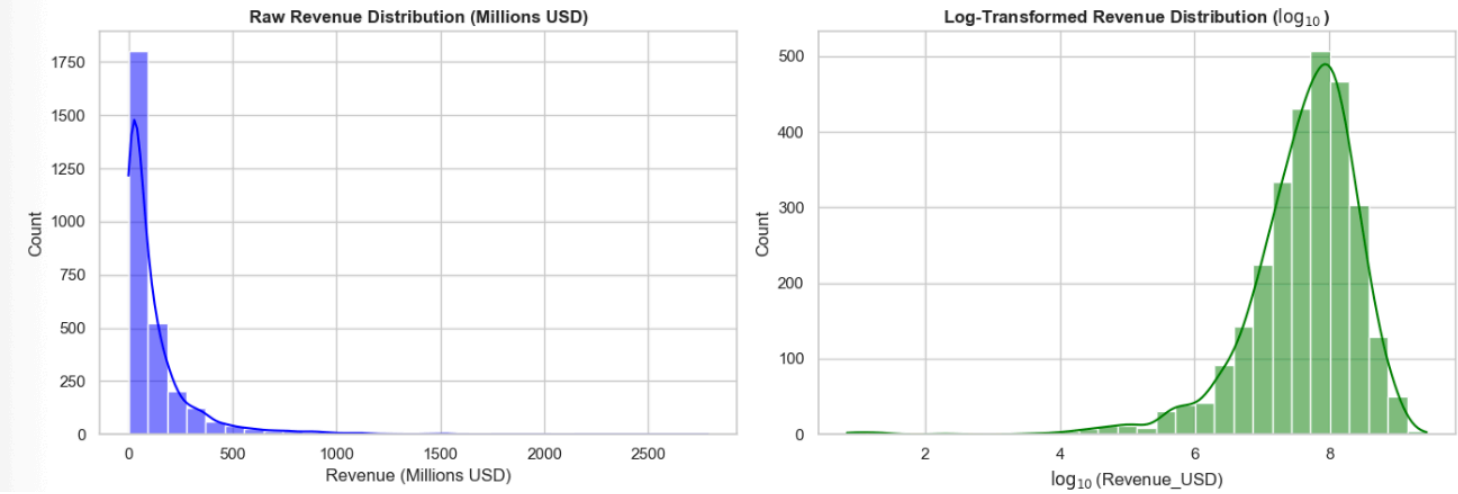
Correlation Matrix of Continuous Features



I then moved to checking the distribution of my target variable, revenue. The distribution of the data is important to understand since Linear Regression will assume that the residuals, the errors or differences between the predicted and actual values, are normally distributed. I plotted two histograms, one with the raw revenue numbers and one with log10 transformed revenue. The raw revenue was heavily right-skewed. The log transformed had a bit more of an even distribution (looking at the bulk of the data), however it still maintained a longer tail showing left-skew. This meant that I would need to train the model on the log transformed revenue instead

Figure 3

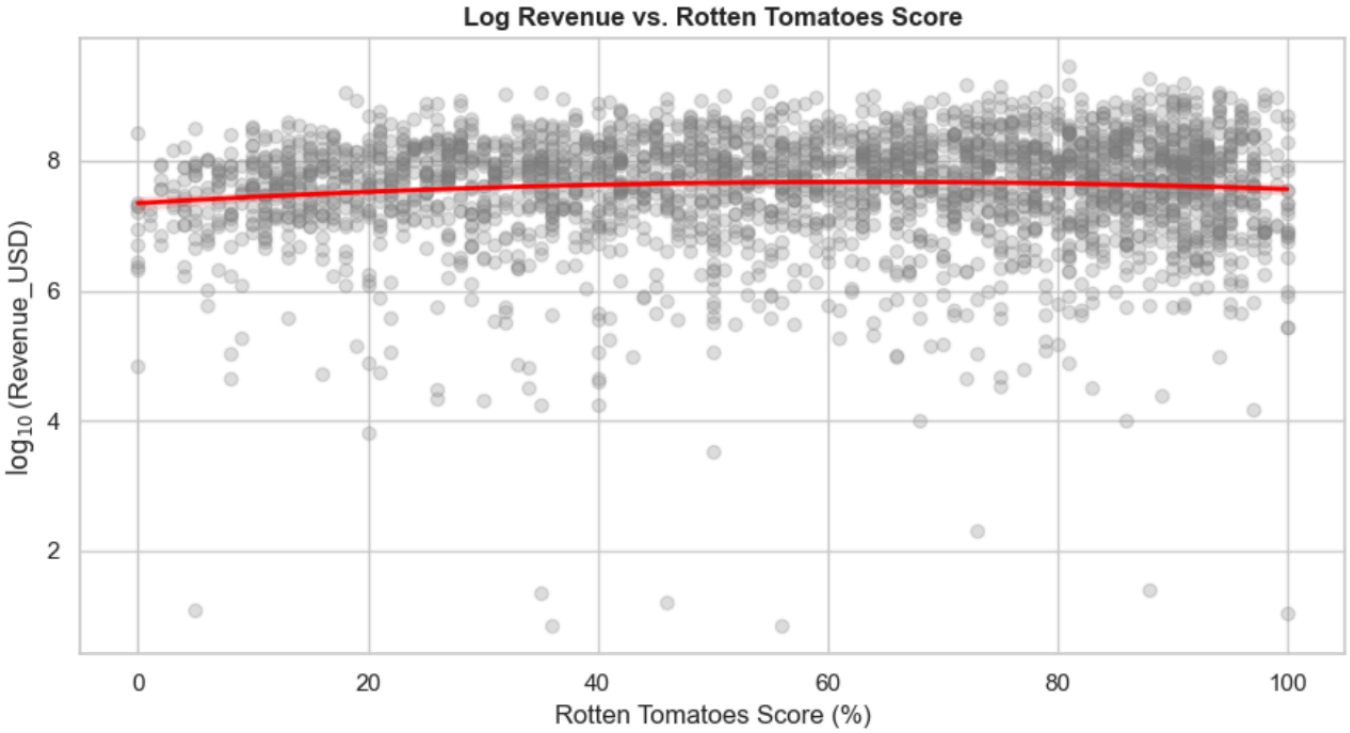
Revenue Distribution



of the raw revenue. Finally, I did a check for linearity of the feature and target variable. I created a scatter plot that overlaid the fitted regression model to evaluate the relationship between *Log Revenue* and the *Rotten Tomatoes Score*. This plotted each individual movie point, then drew the best-fit regression line. Since there is only a very slight curve it indicated a linear assumption was reasonable for this data. A Multiple Linear Regression Model should be a good first choice for this data as it should be able to learn the pattern and perform fairly well.

Figure 4

Log Revenus versus Rotten Tomatoes Score



Results & Findings Explained

Feature Engineering

The correlation heatmap showed that the *Metacritic_Score* and *RT_Score* had a strong positive correlation of 0.92. This is expected since the Rotten tomatoes and the Metacritic Score are both professional ratings. The Metacritic Score comes from the Metacritic site which is an additional professional rating metric that aggregates written reviews from elite, established critics and major publications,

Having two highly correlated features introduces multicollinearity and causes the Multiple Linear Regression model to over-emphasize their importance during testing. Therefore,

to mitigate this without losing valuable data, I engineered a new feature, *Combined_Score,* that combined these two values. This is calculated as the arithmetic mean of the two values:

$$Combined_Score = \frac{Metacritic_Score + RT_Score}{2}$$

I also created a curated subset of features to utilize for modeling: *Budget, Combined Score, IMDb Rating, Popularity Score,* and *Total Votes*. Following Occam's razor, I have done this as a parsimonious, domain-driven baseline, using the simplest set of features possible. This allowed me to keep the interpretability of the models high since these variables are clean, distinct, and easy to explain to stakeholders. I also wanted to avoid having feature importance spread across a number of minor variables which would be harder to interpret. The curated set of values should capture the primary drivers for indications of movie success.

Multiple Linear Regression

Results: R^2: 0.3802 | MAE: $525.26M | RMSE: $5896.79M

The base algorithm for this study was a Multiple Linear Regression, as this would use more than a single independent feature (variable) to predict the target variable of *Revenue*. A Multiple Linear Regression model simultaneously calculates a separate weight, the coefficient, for each of the features in the multi-column DataFrame. Since some movies make a modest amount in the millions, but blockbusters make billions, this creates gaps in the data and makes it harder for a linear regression model to find the patterns in the data. This is why the revenue must be converted to log numbers, which shrinks the gap between small films and the blockbusters.

This makes the residuals have a smoother curve and appear more symmetrical. Training models with log-transformed revenue provides more stability in the math and better predictive power.

Results for the revenue amounts are then given in the log scale, and for them to be interpretable, and to get the raw value amounts, which are easier to understand, the log values must be inverse transformed. Doing this naively gives the median prediction, not the mean. The mean will represent the expected average for all movies, therefore, a smearing correction must be applied to get the mean on the original scale (MetricGate, 2025). Not including the smearing factor is a common mistake during regression analysis. If it is not used then the model calculates the median and pretends it is the mean and will therefore underestimate how much money a movie would be expected to make on average. By utilizing Duan's Smearing Factor it will take the spread of the errors and then calculate exactly how much the outliers (the blockbuster films) add and then multiplies the final prediction by that factor. This was an important consideration for this project because if the median earnings were predicted, instead of the mean earnings, then the studio or production company would consistently underestimate how much revenue they would bring in for their new movies due to ignoring the massive boost from the occasional blockbuster. Reporting in actual dollar amounts ensures clarity and allows for easy comparison across the different models.

Results from this run showed that the model fell into a log scale prediction trap which caused the numbers to appear excessively high and explains why the RMSE was incredibly high at $5896.79M, or $5 Billion, $896.79Million. This was a huge red flag and a mathematical impossibility for a standard movie prediction. For example, the feature film, *Jurassic World: Dominion*, holds the record of being the most expensive film on record with a budget of

$658.8Million (Sylt, 2026). Since the model was trained on log-scale and then inverse-transform back to raw dollars this caused small mistakes to scale to massive proportions. R-Squared showed that the simple straight line using the current features only accounted for 38% of why a movie makes money, thus leaving 62% unaccounted for. This meant that the features had a weak to moderate linear relationship with revenue and that linear data alone was probably not enough and a non-linear tree model would most likely provide more accurate results. The MAE of $525.26M means that, on average, predictions were off by $525.26M which showed the linear model was consistently overestimating or underestimating by large margins. The linear model got confused since high budget flops and low budget sleeper hits caused massive prediction errors.

Optimized Decision Tree Regressor

Results: R^2: 0.6610 | MAE: $64.58M | RMSE: $122.28M

A Decision Tree Regressor is able to handle non-linearity and feature interactions, like how a high budget may yield a high revenue only if it's paired with high audience engagement. Tree models make decisions using step-wise threshold splits, therefore no log transformations were required, and therefore the retransformation bias seen in the Linear model could be avoided. This was a simple first model to evaluate if a non-linear tree model was a better fit for this study. I did some minimal optimizations as a step up from a base Decision Tree model through grid-level optimizations. To prevent the model from overfitting on the training data, or any noise, I performed hyperparameter tuning using a 5-fold cross validation grid search. I set the grid up to train 15 variations of the model which evaluated each run across multiple cross-

validation folds and then selected the best options for the max depth and minimum samples split that yielded the highest R-Squared on the unseen test data. This was a rigorous model for a non-linear baseline comparison to the Multiple Linear Regression.

Results from the Optimized Decision Tree Regressor were more promising than the Multiple Linear Regression. The R-Squared went from 38% to 66% which meant nearly double the variance in the success of a movie could be explained. The average prediction error, MAE, went down by more than 75% from $274.99M to $64.58M. This was a more realistic number for how much, on average, predictions were off by. The RMSE also showed improvements moving from $5,896.79M to $122.28M. Since the Decision Tree used raw revenue amounts rather than log transformed revenue, the smaller errors in predictions didn't transform into larger errors during inverse transformation. The RMSE of $121.84M was more in line with this dataset where the top films were grossing over $2 Billion.

Random Forest Regression

Results: R^2: 0.7252 | MAE: $57.90M | RMSE: $110.10M

Next I tested a Random Forest Regression. It fits an ensemble of multiple independent decision trees on bootstrapped subsets of the data and then averages their predictions. Ensemble methods typically perform well and provide better results since they eliminate the variance issues seen in single decision trees due to the bagging (bootstrap aggregating). Unlike a standard decision tree that will evaluate every feature at every node split, the Random Forest will restrict each tree to a random sample of candidate features. They also handle non-linear feature interactions well and are robust to multicollinearity.

The results from the Random Forest provided even more of a boost than the Optimized Decision Tree, with the R-Squared hitting 72.52%. This was a 6.42% improvement in the R-Squared over the Decision Tree Regressor. This showed how the ensemble averaging reduced individual tree variance and was still able to provide higher predictive accuracy. The average of the absolute differences between the actual and predicted errors, the MAE, decreased even further to $57.90M which cut another $6.68M from the model's average prediction error. The RMSE also decreased to $110.10M. This meant that the model's worst-case prediction errors were off by $110.10M which was another $12.28M cut from the Decision Tree Regressor's worst-case prediction errors.

Since the Random Forest performed the best I wanted to see which of the features were carrying the most weight in predicting the financial success of a movie.

Table 1

Random Forest- Feature Importances

Feature	Importance	Takeaway
Total Votes	58.4%	This was the primary driver of a movie's success, and therefore high revenue was based on votes on movie platforms like IMDb. This is a strong indicator of audience engagement and the volume of viewers. More viewers equates to more revenue. More than half of the model's predictive power came from this single variable.

Feature	Importance	Takeaway
Budget	29.6%	The production budget accounted for just under a third of the model's predictions for determining the success of a film. A high budget will allow for more marketing, better and more movie theaters for release, and bigger movie stars
Popularity	5.6%	This is a very small driver of success and is
Combined Score	3.6%	The average score of Rotten Tomatoes and the Metacritic Score. These only played a minor role in predicting movie success
IMDb Rating	2.8%	Ratings on IMDb played a very small role in the overall success of a movie, but did provide input for any edge cases, such as a sleeper hit or an overhyped dud

These results showed that box office revenue was primarily driven by audience awareness, engagement, and reach of the movie. The more viewers liked the movie and voted for it, the more people were influenced to also go and see the movie, driving up box office revenue. The percentage of importance for votes is nearly double the next highest feature importance, the budget. This meant that a high budget does not dictate the success of a movie, and instead, the movie must generate interest from viewers so they will vote for it. The remaining features were merely minor refinements for the model to help gauge how sleeper hits or high-budget flops would perform and the amount of revenue they would generate.

Table 2

Base Model Comparison

Model Type	R-Squared	MAE	RMSE
Multiple Linear Regression	38.02%	$525.26M	$5,896.79M
Decision Tree Regressor	66.10%	$64.58M	$122.28M
Random Forest	72.52%	$57.90M	$110.10M

Random Forest Regression with Tuning and Optimizations

Results: R^2: 0.7393 | MAE: $58.54M | RMSE: $107.25M

Optimal Parameters

Max Depth: 10 | Max Features: 0.4 | Minimum Samples Leaf: 1 | Minimum Samples Split: 2 | N Estimators: 500

Duan Smearing Factor: 1.2965

To test if there was any better performance I could get from the Random Forest Regression I did some tuning with the randomized search to sample broadly across the wide parameter distributions, and then performed a narrow grid search, which is more fine-grained, and centers around the best parameters found during the randomized search. Since a Random Forest will minimize prediction errors by evaluating every decision split to reduce loss, if a blockbuster and indie film had prediction errors of $400 Million and $9 Million respectively, the model would square these errors making the blockbuster film's mistake nearly 2,000 times larger

to the model. The model would then ignore the indie film and would spend all of its tree splits attempting to fix the prediction error from the blockbuster film.

This reasoning is why I log transformed some of the features. Log Transformation of the target variable, *Revenue,* would prevent the Random Forest Regressor from being heavily biased toward the mega-blockbuster movies by training the model on relative percentage errors, rather than raw dollar differences. Log Transformations of the features, *Budget* and *Total Votes* compressed the right skewed predictors and enabled individual decision trees to isolate patterns to make more granular splits across the lower and medium budget films. I then added a 5-fold cross-validation search grid that enabled the model to evaluate 192 parameter combinations across the tree count, max tree depth, feature subspace sampling, and leaf node split constraints. This lead to a total of 960 different model fit evaluations from which the single best performer was extracted to be used for the final test set evaluation.

I included additional engineered features to provide more explicit signals for the trees to split on, instead of focusing on the total votes or budget. First, to measure marketing and financial efficiency, versus the raw audience engagement numbers, I created a *Budget Per Vote* feature that divided the budget by total votes.

$$Budget_Per_Vote = \frac{Budget}{Total_Votes + 1}$$

I was hoping to capture the affect that the budget had on the votes for a film, and if I could see if a specific budget value amount, based on the votes received, translated to film success.
Then, to capture the difference between how critics vote, seen in the *Combined Score*, and the general population votes, the IMDb Ratings, I created a *Critic User Diff*, or the Critic user

differential. As the *Combined Score* is a percentage from 0-100, and the IMDb Rating is a 1-10 number; I will multiply the IMDb Rating by 10 to align both values on the same mathematical scale.

$$Critic_User_Diff = Combined_Score - (IMDb_Rating \times 10)$$

Here I was trying to measure consumer polarization and market sentiment alignment. I was hoping to evaluate whether movies where the critics and the general public's votes *agreeing* meant a film would perform with more predictability over films who measured stronger either with the critics or the public.

I also created additional adjustments by increasing trees to 300 and 500 to help reduce variance without increasing the risk of overfitting. By fine tuning the `max_features` the model would evaluate fewer features per split and be forced to build trees around secondary indicators instead of doing greedy splits on the *Total Votes*. Lower minimum samples per leaf would also allow individual trees to isolate tight clusters of high performing films.

All of these optimizations led to a small performance increase. The R-Squared increased to 73.9%. The MAE increased slightly to $58.54 Million, which was a slight increase in the average magnitude of absolute errors in the predictions. However, the RMSE saw a modest reduction to $107.25 Million, which meant the model's errors on the outlier films saw a $2.85 Million reduction in error prediction from the Random Forest Regressor. This shows that the outliers are difficult for the model to perform accurately on, since RMSE is sensitive to large errors, which is shown in this score.

The feature importance breakdown saw a massive shift. Due to the newly engineered features, and utilizing log scale for the budget and the votes, the votes were no longer used as the

biggest predictor for film success. Instead, here, the *log budget* and *log votes* features are much closer tied at 26.86% and 26.33% respectively. Followed by the *Popularity Score* at 20.95%.

Table 3

Random Forest Regression with Tuning and Optimizations

Feature	Importance	Takeaway
Log Budget	26.86%	Budget was now the highest predictor of a movie's financial success, but just narrowly so.
Log Votes	26.33%	Log Votes was a very close runner up to the Log Budget. Between the Log Budget and Log Votes, these two features accounted for just over 50% of the model's predictive power.
Popularity Score	20.95%	The Popularity Score was now nearly a quarter of the model's predictive power. This showed that between the votes and popularity score, the amount of audience engagement and interactions are huge predictors of a movie's financial success. These two combined features account for 47.28% of predicting the success of a movie.
Budget Per Vote	13.88%	Combining the Log Budget and Budget Per Vote, these two features accounted for 40.76% of determining the success of a film.
Combined Score	4.08%	The average score of Rotten Tomatoes and the Metacritic Score. These only played a minor role in predicting movie success

Feature	Importance	Takeaway
Critic User Diff	3.97%	This feature is capturing the difference between the Combined Score, which comes from the professional movie critics and the average person. This was used to capture those movies that were 'duds' to the professional critics but the general audiences loved.
IMDb Rating	3.95%	Ratings on IMDb play a very small role in the overall success of a movie, but did provide input for any edge cases such as a sleeper hit or an overhyped dud

Conclusion

The **Random Forest Regressor'**s feature importance showed that the most important feature in predicting the revenue for a film was *Total Votes*, with *Budget* being the next runner up at only about half of the percentage amount of the *Total Votes*. In the **Optimized Random Forest Regressor**, the *Log Budget* was nearly tied with *Log Votes* as the top features for predicting the revenue of a movie. The un-transformed target variance in the Random Forest Regressor caused the model to over-index on the raw audience volume which resulted in the higher feature importance of the *Total Votes*. Once the target was log transformed and hyperparameter optimization was employed, the model's reliance on the extreme vote counts was effectively mitigated. This led to the *Log Budget* emerging as the top individual predictor followed very closely by *Log Votes.* The newly engineered feature of *Budget per Vote* pushed the total financial allocation to slightly over 40%. However, the combination of *Log Votes* and *Popularity Score* account for 47.28% of the score in predicting the financial success of a film.

This shows that audience engagement through voting, marking the film as a favorite, or adding to their watchlist on TMDB or voting for the film on IMDb drives the success of a movie even more than how much was spent to produce it. The feature importance analysis revealed that sentiment divergence between critics and the general public accounted for less than 4% of model variance. This confirmed that movie success is driven by budget and mass market reach to the general public rather than reception by movie critics.

The final optimized Random Forest Regressor provided the best results in this study, with the highest R-Squared and lowest RMSE. The R-Squared of 73.93% shows the independent variables are able to explain the variability of the dependent target variable nearly three quarters of the time. The MAE is the average of the absolute differences between the actual and predicted values. This showed us how far the model's predictions are off in dollar amounts. The MAE of $58.54 Million shows the average predictions were off by about $58 Million. For Mega Blockbuster films, who cost $200- $450 Million this is a fraction of their budget. For Studio films, these range from $50 - $150 Million, makes this prediction error seem larger. Due to the wide variety of budget values, with 7 titles listed under $30 for budget (which seems unlikely), to $380 Million, this error is understandable. This distribution made it difficult for the model to predict with closer accuracy. To get better model accuracy I would likely need to trim out more of the unreasonably low sounding budgets or look to find another data source that has better metrics for production budgets.

Table 4

Final Model Results

Model Type	R-Squared	MAE	RMSE
Multiple Linear Regression	38.02%	$525.26M	$5,896.79M
Decision Tree Regressor	66.10%	$64.58M	$122.28M
Random Forest	72.52%	$57.90M	$110.10M
Optimized Random Forest	73.93%	$58.54M	$107.25M

Recommendations

The Optimized Random Forest model would be a good choice to run to predict post-release revenue forecasting, once initial general public votes have been made. The results would need to be reviewed through a critical lens and treated as merely a tentative projection of the movie's revenue. Due to the MAE of $58.54M, the average magnitude of absolute errors in prediction for how much a movie will make was off by $58.54M; for a smaller movie this is a massive error, but for a blockbuster, this is an acceptable risk. The RMSE of $107.25M means this model is suffering from large misses on breakout hit movies. The business impact from these results means if a low-budget movie, say $5M, is a huge success and makes $250M then the model will grossly underestimate its revenue. Conversely, if a film with a massive budget, say $200M, flops and only makes $40M, then the model will incorrectly predict a high revenue and be severely off. For the average movie, the model can be considered more reliable than the others

attempted in this study but it should not be trusted for high-stakes financial forecasting due to how severe the worst-case errors are.

This study focused on evaluating the predictive relationship between a movie's budget, its reception metrics, from both critics and the general public, and its revenue to determine if a higher budget alone would dictate financial success of a film. By combining data from both the Movie Database (TMDB) and the Open Movie Database (OBDb) API, I was able to create a unified dataset of nearly 3,000 titles for my analysis and modeling.

The findings from the different models utilized showed that budget alone is not enough to predict if a film will be a financial success. The initial baseline linear model proved to be inadequate at capturing the more complex, non-linear financial dynamics of film creation. However, the ensemble tree-based models were more able to successfully isolate the key drivers of movie success. The final optimizations completed on the Random Forest Regressor, through hyperparameter tuning and log-transformed variables was able to explain 73.93% of the variance in box office revenue. The feature importance analysis revealed that, while *Log Budget,* at 26.86%, proved necessary for production and distribution, the engagement by audiences through the *Log Votes, Popularity Score,* and *Budget Per Vote,* collectively accounted for over 61% of the feature importance and represents the true prediction of movie success. The low importance of critic metrics and critic-user sentiment differential confirmed that general public engagement held a significantly greater influence than professional critic reviews.

Operationally, these insights can offer production studios and filmmakers a data-driven approach to risk assessment. However, there are critical limitations that must be considered. The final model provided a Mean Absolute Error (MAE) of $58.54M and a Root Mean Squared Error

(RMSE) of $107.25M, which is primarily driven by the extreme variance seen in box office hits and low-budget successes. Consequently, this model can only serve as a reliable tool for a mid-tier studio release with post-release tracking. For any high-stakes decisions by a production company to invest a higher budget into creation of a film would require human oversight. This project could be expanded upon, in a number of ways, to attain more accurate results. First, the films could be broken up by spending tiers to bucket production budgets into smaller groups so there is not such a wide swing for the model to learn. This would help to limit the large prediction errors seen during modeling. Then, additional time-series features, such as examining pre-release marketing expenditure, opening weekend revenue totals, and social media sentiment analysis could help capture audience response before the movie's release.

References

Chugh, A. (2020) *MAE, MSE, RMSE, coefficient of determination, adjusted r squared - which metric is better?* Medium. https://medium.com/analytics-vidhya/mae-mse-rmse-coefficient-of-determination-adjusted-r-squared-which-metric-is-better-cd0326a5697e

Cinevue. (2020). *The true cost of movie-making.* Cinevue. https://cine-vue.com/2020/11/the-true-cost-of-movie-making.html

Fritz, B. (2026). *OMDb API* (Version 1) [Computer software]. https://www.omdbapi.com/

MetricGate. (2025). *Log transformations for skewed data.* MetricGate. https://metricgate.com/blogs/log-transformations-skewed-data/

OMDb API. (n.d.) *OMDb API The open movie database.* OMDb API. https://www.omdbapi.com/

OMDb API. (n.d.) *API Key.* OMDb API. https://www.omdbapi.com/apikey.aspx

Sylt, C. (2026). *Exclusive: universal beat Disney as Hollywood's maker of the most expensive movie of all time.* Fortune. https://fortune.com/2026/06/17/most-expensive-movie-ever-jurassic-world-dominion-beating-star-wars-the-force-awakens/

The Movie Database. (2017). *TMDB 5000 Movie Dataset* (Version 2) [Data set]. Kaggle. https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata